

GENERAL INFORMATION

DIAGNOSTIC TROUBLE CODE INDEX - DTC:
DRIVER/PASSENGER SEAT MODULE [DSM/PSM]

[G2006793]

DESCRIPTION AND OPERATION

DRIVER/PASSENGER SEAT MODULE [DSM/PSM]

CAUTION:

Diagnosis by substitution from a donor vehicle is **NOT** acceptable. Substitution of control modules does not guarantee confirmation of a fault, and may also cause additional faults in the vehicle being tested and/or the donor vehicle

NOTES:

- Guided Diagnostics must be followed and the repair advised by the process completed. Failure to do so may result in the rejection of any warranty claim made.
- If the control module or a component is at fault and the vehicle remains under manufacturer warranty, refer to the Warranty Policy and Procedures manual, or determine if any prior approval program is in operation, prior to the installation of a new module/component
- Generic scan tools may not read the codes listed, or may read only 5-digit codes. Match the 5 digits from the scan tool to the first 5 digits of the 7-digit code listed to identify the fault (the last 2 digits give extra information read by the manufacturer-approved diagnostic system)
- When performing voltage or resistance tests, always use a digital multimeter accurate to three decimal places and with a current calibration certificate. When testing resistance, always take the resistance of the digital multimeter leads into account
- Check and rectify basic faults before beginning diagnostic routines involving pinpoint tests
- Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
- If diagnostic trouble codes are recorded and, after performing the pinpoint tests, a fault is not present, an intermittent concern may be the cause. Always check for loose connections and corroded terminals
- Where an 'on demand self-test' is referred to, this can be accessed through the 'diagnostic trouble code monitor' tab on the manufacturers approved diagnostic system
- Check JLR claims submission system for open campaigns. Refer to the corresponding bulletins and SSMs which may be valid for the specific customer complaint and complete the recommendations as required.

The table below lists all diagnostic trouble codes (DTCs) that could be set in the driver/passenger seat module, for additional diagnosis and testing information refer to the relevant diagnosis and testing section. For additional information, refer to: Seats (501-10 Seating, Diagnosis and Testing).

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
B1087-88	LIN Bus A - Bus off	NOTE: Circuit reference - LIN1 LIN bus circuit short circuit to ground, short circuit to power, open circuit, high resistance	Refer to the electrical circuit diagrams and check the LIN bus circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary
B12CC-00	Driver Seat Massage Module Pump Assembly - No sub type information	- Driver seat massage air pump circuit open circuit, high resistance - Driver seat massage air pump pneumatic hose leaking	NOTE: This DTC can only be set if the fault is detected during the Massage ECU On Demand Self Test routine. - Refer to the electrical circuit diagrams and check the driver seat massage air pump circuit for open circuit, high resistance. Repair the wiring harness or install a new massage air pump as necessary - Check the driver seat massage air pump pneumatic hoses for leaks. Rectify as necessary
B12D9-00	Driver Seat Massage Module Back Carrier Assembly - No sub type information	- Driver seat massage slave solenoid circuit open circuit, high resistance - Driver seat massage slave solenoid pneumatic hose leaking or restricted	NOTE: This DTC can only be set if the fault is detected during the Massage ECU On Demand Self Test routine. - Refer to the electrical circuit diagrams and check the driver seat massage slave solenoid circuit for open circuit, high resistance. Repair the wiring harness or install a new massage slave solenoid as necessary - Check the driver seat massage slave solenoid pneumatic hoses for leaks and restrictions. Rectify as necessary
B12DB-00	Passenger Seat Massage Module Pump Assembly - No sub type information	- Passenger seat massage air pump circuit open circuit, high resistance - Passenger seat massage air pump pneumatic hose leaking	NOTE: This DTC can only be set if the fault is detected during the Massage ECU On Demand Self Test routine. - Refer to the electrical circuit diagrams and check the passenger seat massage air pump circuit for open circuit, high resistance. Repair the wiring harness or install a new massage air pump as necessary - Check the passenger seat massage air pump pneumatic hoses for leaks. Rectify as necessary

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
B12E6-00	Passenger Seat Massage Module Back Carrier Assembly - No sub type information	- Passenger seat massage slave solenoid circuit open circuit, high resistance - Passenger seat massage slave solenoid pneumatic hose leaking or restricted	NOTE: This DTC can only be set if the fault is detected during the Massage ECU On Demand Self Test routine. - Refer to the electrical circuit diagrams and check the passenger seat massage slave solenoid circuit for open circuit, high resistance. Repair the wiring harness or install a new massage slave solenoid as necessary - Check the passenger seat massage slave solenoid pneumatic hoses for leaks and restrictions. Rectify as necessary
B14EF-18	Bank 'A' Motor #1 Output - Circuit current below threshold	12-way specification seating: Seat height motor circuit open circuit, high resistance	12-way specification seating: Refer to the electrical circuit diagrams and check the seat height motor circuit for open circuit, high resistance. Repair the wiring harness or install a new seat height motor as necessary
B14EF-19	Bank 'A' Motor #1 Output - Circuit current above threshold	12-way specification seating: Seat height motor circuit short circuit to ground, short circuit to power	12-way specification seating: Refer to the electrical circuit diagrams and check the seat height motor circuit for short circuit to ground, short circuit to power. Repair the wiring harness or install a new seat height motor as necessary
B14F0-18	Bank 'A' Motor #2 Output - Circuit current below threshold	6-way specification seating: Seat forward/rearward motor circuit open circuit, high resistance	6-way specification seating: Refer to the electrical circuit diagrams and check the seat forward/rearward motor circuit for open circuit, high resistance. Repair the wiring harness or install a new seat forward/rearward motor as necessary
B14F0-19	Bank 'A' Motor #2 Output - Circuit current above threshold	6-way specification seating: Seat forward/rearward motor circuit short circuit to ground, short circuit to power	6-way specification seating: Refer to the electrical circuit diagrams and check the seat forward/rearward motor circuit for short circuit to ground, short circuit to power. Repair the wiring harness or install a new seat forward/rearward motor as necessary
B14F1-18	Bank 'A' Motor #3 Output - Circuit current below threshold	6-way specification seating: Seat height motor circuit open circuit, high resistance 12-way specification seating: Seat forward/rearward motor circuit open circuit, high resistance	6-way specification seating: Refer to the electrical circuit diagrams and check the seat height motor circuit for open circuit, high resistance. Repair the wiring harness or install a new seat height motor as necessary 12-way specification seating: Refer to the electrical circuit diagrams and check the seat forward/rearward motor circuit for open circuit, high resistance. Repair the wiring harness or install a new seat forward/rearward motor as necessary
B14F1-19	Bank 'A' Motor #3 Output - Circuit current above threshold	6-way specification seating: Seat height motor circuit short circuit to ground, short circuit to power 12-way specification seating: Seat forward/rearward motor circuit short circuit to ground, short circuit to power	6-way specification seating: Refer to the electrical circuit diagrams and check the seat height motor circuit for short circuit to ground, short circuit to power. Repair the wiring harness or install a new seat height motor as necessary 12-way specification seating: Refer to the electrical circuit diagrams and check the seat forward/rearward motor circuit for short circuit to ground, short circuit to power. Repair the wiring harness or install a new seat forward/rearward motor as necessary

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
B14F3-18	Bank 'B' Motor #1 Output - Circuit current below threshold	12-way specification seating: Seat cushion tilt motor circuit open circuit, high resistance	12-way specification seating: Refer to the electrical circuit diagrams and check the seat cushion tilt motor circuit for open circuit, high resistance. Repair the wiring harness or install a new seat cushion tilt motor as necessary
B14F3-19	Bank 'B' Motor #1 Output - Circuit current above threshold	12-way specification seating: Seat cushion tilt motor circuit short circuit to ground, short circuit to power	12-way specification seating: Refer to the electrical circuit diagrams and check the seat cushion tilt motor circuit for short circuit to ground, short circuit to power. Repair the wiring harness or install a new seat cushion tilt motor as necessary
B14F4-18	Bank 'B' Motor #2 Output - Circuit current below threshold	Seat backrest release motor circuit open circuit, high resistance	Refer to the electrical circuit diagrams and check the seat backrest release motor circuit for open circuit, high resistance. Repair the wiring harness or install a new seat backrest release motor as necessary
B14F4-19	Bank 'B' Motor #2 Output - Circuit current above threshold	Seat backrest release motor circuit short circuit to ground, short circuit to power	Refer to the electrical circuit diagrams and check the seat backrest release motor circuit for short circuit to ground, short circuit to power. Repair the wiring harness or install a new seat backrest release motor as necessary
B14FB-31	Bank 'A' Motor #1 Speed/Position Sensor - No signal	12-way specification seating: Seat height motor position sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance	12-way specification seating: Refer to the electrical circuit diagrams and check the seat height motor position sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness or install a new seat height motor as necessary
B14FC-31	Bank 'A' Motor #2 Speed/Position Sensor - No signal	12-way specification seating: Seat forward/rearward motor position sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance	12-way specification seating: Refer to the electrical circuit diagrams and check the seat forward/rearward motor position sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness or install a new seat forward/rearward motor as necessary
B14FD-31	Bank 'A' Motor #3 Speed/Position Sensor - No signal	6-way specification seating: Seat height motor position sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance 12-way specification seating: Seat forward/rearward motor position sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance	6-way specification seating: Refer to the electrical circuit diagrams and check the seat height motor position sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness or install a new seat height motor as necessary 12-way specification seating: Refer to the electrical circuit diagrams and check the seat forward/rearward motor position sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness or install a new seat forward/rearward motor as necessary
B14FF-31	Bank 'B' Motor #1 Speed/Position Sensor - No signal	12-way specification seating: Seat cushion tilt motor position sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance	12-way specification seating: Refer to the electrical circuit diagrams and check the seat cushion tilt motor position sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness or install a new seat cushion tilt motor as necessary

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
B1500-31	Bank 'B' Motor #2 Speed/Position Sensor - No signal	Seat backrest recline motor position sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance	Refer to the electrical circuit diagrams and check the seat backrest recline motor position sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness or install a new seat backrest recline motor as necessary
B1513-18	5A Battery Side Output Drive #1 - Circuit current below threshold	Lumbar air pump (high) circuit short circuit to ground, short circuit to power	Refer to the electrical circuit diagrams and check the rear seat lumbar air pump (high) circuit for short circuit to ground, short circuit to power. Repair circuit as required. Clear the DTCs and retest
B1513-19	5A Battery Side Output Drive #1 - Circuit current above threshold	Lumbar air pump (high) circuit short circuit to ground, short circuit to power	Refer to the electrical circuit diagrams and check the rear seat lumbar air pump (high) circuit for short circuit to ground, short circuit to power. Repair circuit as required. Clear the DTCs and retest
B1514-18	5A Battery Side Output Drive #2 - Circuit current below threshold	Lumbar air pump (high) circuit short circuit to ground, short circuit to power	Refer to the electrical circuit diagrams and check the rear seat lumbar air pump (high) circuit for short circuit to ground, short circuit to power. Repair circuit as required. Clear the DTCs and retest
B1514-19	5A Battery Side Output Drive #2 - Circuit current above threshold	Lumbar air pump (high) circuit short circuit to ground, short circuit to power	Refer to the electrical circuit diagrams and check the rear seat lumbar air pump (high) circuit for short circuit to ground, short circuit to power. Repair circuit as required. Clear the DTCs and retest
B1515-18	3A Battery Side Output Drive #1 - Circuit current below threshold	Lumbar air pump (low) circuit short circuit to ground, short circuit to power	Refer to the electrical circuit diagrams and check the rear seat lumbar air pump (low) circuit for short circuit to ground, short circuit to power. Repair circuit as required. Clear the DTCs and retest
B1515-19	3A Battery Side Output Drive #1 - Circuit current above threshold	Lumbar air pump (low) circuit short circuit to ground, short circuit to power	Refer to the electrical circuit diagrams and check the rear seat lumbar air pump (low) circuit for short circuit to ground, short circuit to power. Repair circuit as required. Clear the DTCs and retest
B1516-18	3A Battery Side Output Drive #2 - Circuit current below threshold	Lumbar air pump (low) circuit short circuit to ground, short circuit to power	Refer to the electrical circuit diagrams and check the rear seat lumbar air pump (low) circuit for short circuit to ground, short circuit to power. Repair circuit as required. Clear the DTCs and retest
B1516-19	3A Battery Side Output Drive #2 - Circuit current above threshold	Lumbar air pump (low) circuit short circuit to ground, short circuit to power	Refer to the electrical circuit diagrams and check the rear seat lumbar air pump (low) circuit for short circuit to ground, short circuit to power. Repair circuit as required. Clear the DTCs and retest
B1543-24	Front Seat Switch Pack - Signal stuck high	Seat switchpack switch stuck active	NOTE: This DTC may be induced by the driver operating a seat switchpack switch for longer than 120 seconds. Check the operation of the seat switchpack switches. Free any stuck switches or install a new seat switchpack as necessary

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
B1543-87	Front Seat Switch Pack - Missing message	■ Seat switchpack power or ground circuit open circuit, high resistance ■ LIN bus circuit short circuit to ground, short circuit to power, open circuit, high resistance ■ Seat switchpack internal failure	■ Refer to the electrical circuit diagrams and check the seat switchpack power and ground circuits for open circuit, high resistance. Repair the wiring harness as necessary ■ Refer to the electrical circuit diagrams and check the LIN bus circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary ■ Using the Jaguar Land Rover approved diagnostic equipment, clear the DTCs and retest. If the fault persists, install a new seat switchpack
B1544-24	Front Seat Memory Switch Pack - Signal stuck high	■ Seat memory switchpack switch stuck active	NOTE: This DTC may be induced by the driver operating a seat memory switchpack switch for longer than 120 seconds. ■ Check the operation of the seat memory switchpack switches. Free any stuck switches or install a new seat memory switchpack as necessary
B1545-87	Front Seat Massage Module - Missing message	■ Massage slave solenoid circuit short circuit to ground, short circuit to power, open circuit, high resistance	■ Refer to the electrical circuit diagrams and check the massage slave solenoid circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness or install a new massage slave solenoid as necessary
U0037-87	Vehicle Communication Bus B - Missing message	■ Missing message from another control module through the high speed CAN bus (body)	■ Using the Jaguar Land Rover approved diagnostic equipment, check the snapshot data to determine the missing message source control module. Check the relevant control module for related DTCs and refer to the relevant DTC index
U0037-88	Vehicle Communication Bus B - Bus off	■ High speed CAN bus (body) circuit short circuit to ground, short circuit to power, open circuit, high resistance	■ Using the Jaguar Land Rover approved diagnostic equipment, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the high speed CAN bus (body) circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary
U1000-00	Solid State Driver Protection Active - Driver Disabled - No sub type information	■ seat module output circuit short circuit to ground, short circuit to power	NOTE: This DTC is set when one or more short circuits have been detected. The relevant output circuits have been disabled to protect the seat module from permanent damage. ■ Using the Jaguar Land Rover approved diagnostic equipment, check for other short circuit related DTCs and perform the relevant corrective actions. To re-enable the output circuits, using the Jaguar Land Rover approved diagnostic equipment, perform routine - Reset Short Protection
U1A4C-00	Build / End of Line Mode Active - No sub type information	■ Seat module is not configured correctly	■ Using the Jaguar Land Rover approved diagnostic equipment, configure the seat module as a new module

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
U2300-54	Central Configuration - Missing calibration	Car configuration file mismatch with vehicle specification	NOTE: After updating the car configuration file, set the ignition to on and wait 30 seconds before clearing the DTCs. Using the Jaguar Land Rover approved diagnostic equipment, check and up-date the car configuration file as necessary. Clear the DTCs and retest
U2300-55	Central Configuration - Not configured	Car configuration file mismatch with vehicle specification	NOTE: After updating the car configuration file, set the ignition to on and wait 30 seconds before clearing the DTCs. Using the Jaguar Land Rover approved diagnostic equipment, check and up-date the car configuration file as necessary. Clear the DTCs and retest
U2300-56	Central Configuration - Invalid/incomplete configuration	Car configuration file mismatch with vehicle specification	NOTE: After updating the car configuration file, set the ignition to on and wait 30 seconds before clearing the DTCs. Using the Jaguar Land Rover approved diagnostic equipment, check and up-date the car configuration file as necessary. Clear the DTCs and retest
U3000-49	Control Module - Internal electronic failure	- Short circuit protection - DTC U1000-00 - was activated too many times - Seat module internal failure	NOTES: - DTC cannot be cleared. - Using the Jaguar Land Rover approved diagnostic equipment, check for other short circuit related DTCs and perform the relevant corrective actions first. - Using the Jaguar Land Rover approved diagnostic equipment, install a new seat module as necessary
U3000-76	Control Module - Wrong mounting position	Seat module has been configured to a specific seat location and has subsequently been removed and re-installed to a different seat	Check that the seat module is installed to the correct seat. Rectify as necessary

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
U3001-46	Control Module Improper Shutdown - Calibration/parameter memory failure	■ Seat operation interrupted by power disconnection	NOTE: Do NOT replace the module to repair this DTC ■ Using the Jaguar Land Rover approved diagnostic equipment, check the snapshot data (0x4237) to identify the lost calibration axes ■ If the front seat axes have lost calibration, using the Jaguar Land Rover approved diagnostic equipment, perform routine - Calibration Run (0x205F). Clear the DTCs and retest ■ If the rear seat axes have lost calibration, using the Jaguar Land Rover approved diagnostic equipment, perform routine - Calibration Run (0x2055). Clear the DTCs and retest
U3002-81	Vehicle Identification Number - Invalid serial data received	■ Seat module previously installed on another vehicle ■ New seat module installed and VIN not yet programed	■ Install the original or a new seat module as necessary ■ Using the Jaguar Land Rover approved diagnostic equipment, configure the seat module as a new module